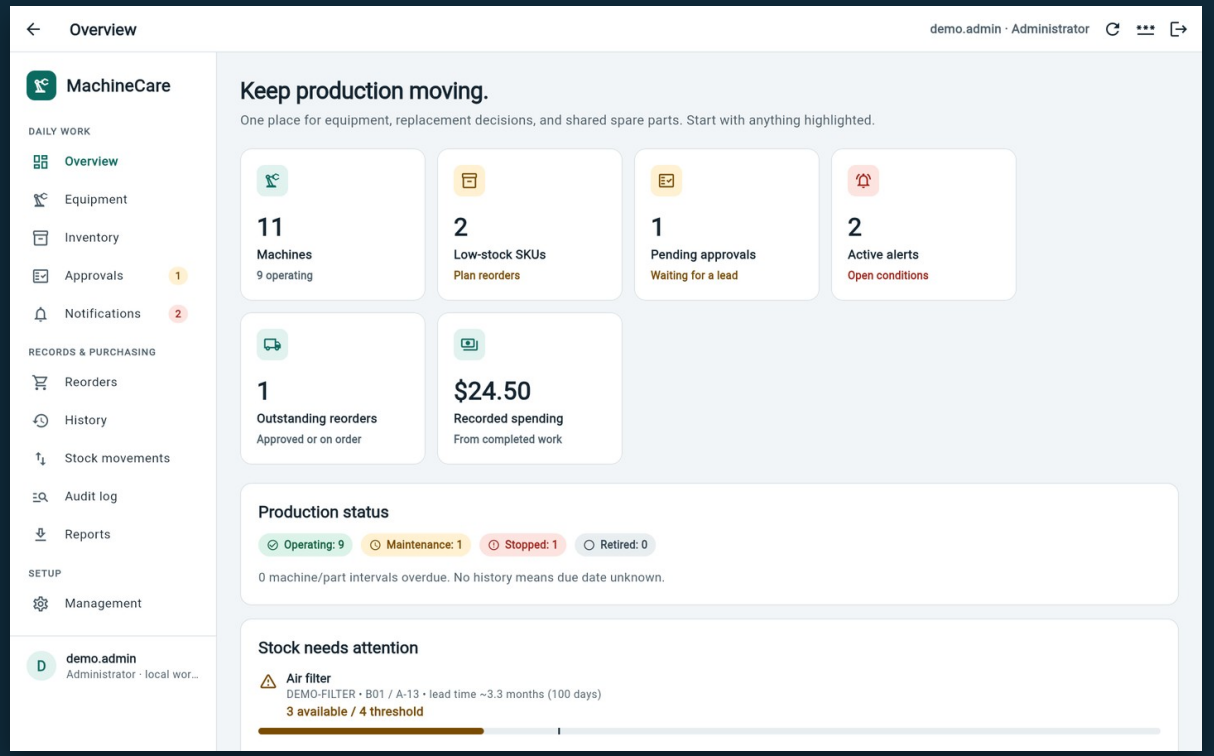
MANAGER REVIEW · 19 OCTOBER 2026

MachineCare

Equipment, spare parts and every replacement in one traceable record

- Working local pilot for DIMM press and AOI lines
- Walkthrough of the app as it runs today
- What still stands between the pilot and production

Ask: approve a one-line validation pilot



WHY THIS MATTERS

Three questions maintenance must answer quickly

Which machine?

Building › line › station › machine, with the parts each machine type is compatible with.

What do we have?

Physical, reserved and available stock for shared spares, and low-stock alerts before a line waits on a part.

What happened?

Who replaced which part, when, where, why, and at what cost, recorded at the moment the work is completed.

One workflow: Request → Review → Reserve stock → Complete → Traceable history

WHAT EXISTS TODAY

A complete working pilot, not a mock-up

11

screens

Overview to Reports

3

roles

Admin · Lead · Technician

11

machines

6 DIMM + 3 AOI + 2 others

20

catalog parts

with real DIMM/AOI references

Built with

- Flutter web front end (works on desktop, tablet, phone)
- Python API using only the standard library
- SQLite database with 7 ordered schema migrations
- Optional background worker: checks, backups, email

Verified by automated tests

- 140 backend tests: permissions, stock, workflows, security, backups
- 45 Flutter widget tests: screens, roles, forms, layouts
- Static analysis: no issues found

PRODUCTION MODEL

Every configured line: two DIMM presses and one AOI



10
DIMM part references

7
AOI component references

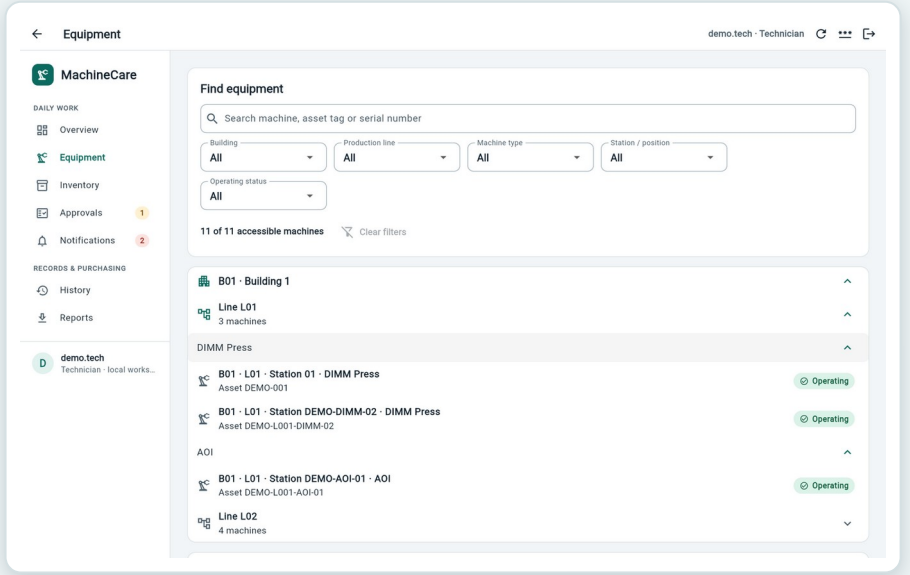
6 DIMM + 3 AOI
+ existing ACOS station and conveyor = 11 equipment records

Part names, manufacturers, model numbers and quantities per machine were supplied by maintenance (see appendix). New asset tags, stations and statuses are placeholders until a physical survey.

WHO DOES WHAT

Three roles, enforced on the server

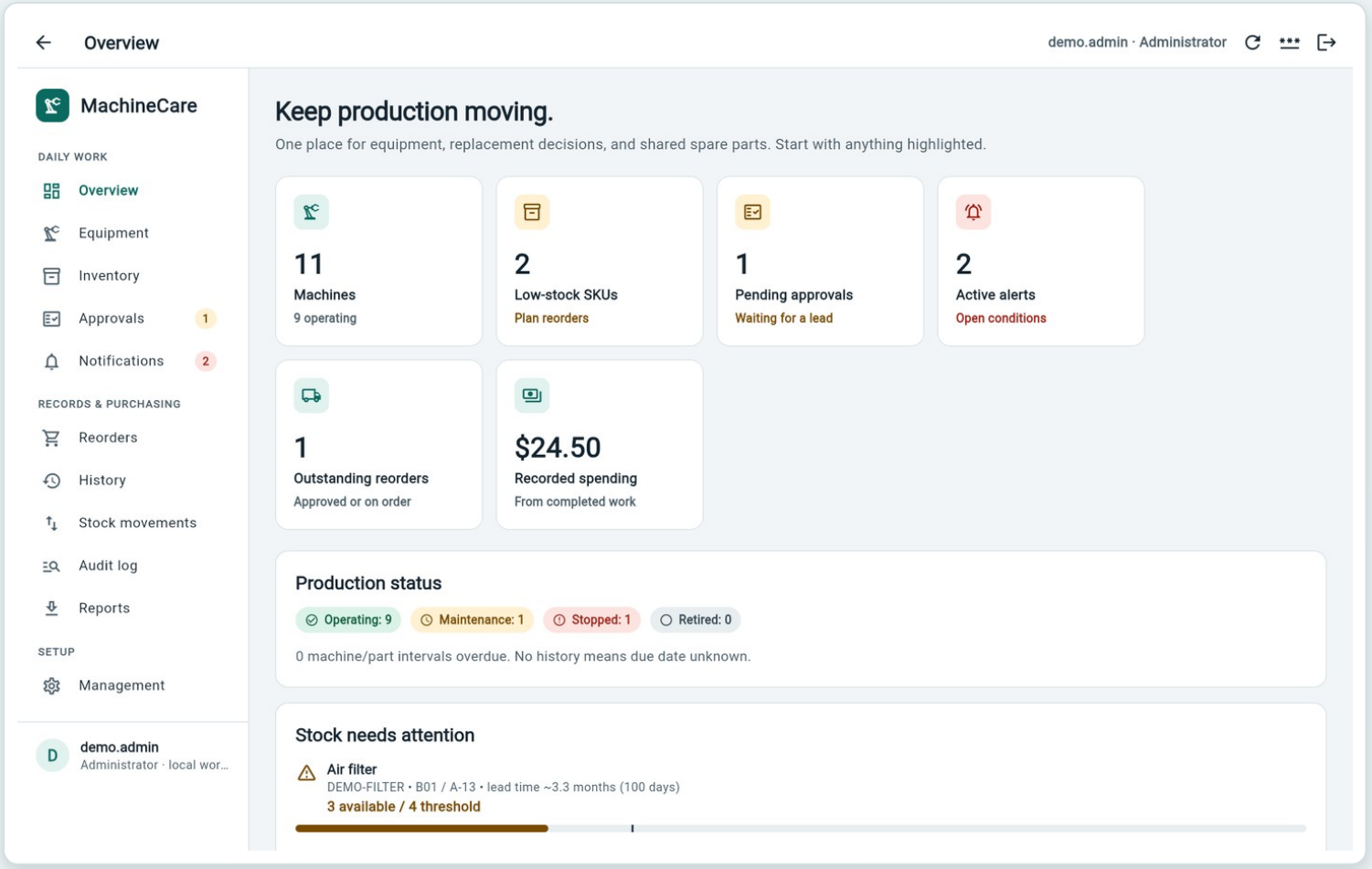
Capability	Admin	Lead	Tech
View assigned machines and history	✓	✓	✓
Request a replacement; complete own work	✓	✓	✓
Review early work, complete others' work	✓	✓	–
Catalog, stock receipts and adjustments	✓	✓	–
Purchasing, audit log, stock ledger, exports	✓	✓	–
Accounts, access, buildings and lines, backups	✓	–	–



Technician view: no Management, Reorders, Audit or Stock screens

Technicians see a shorter menu, and only the lines they are assigned to. Hiding buttons is not the protection: the server checks every action.

The overview puts today's exceptions first



Start with what needs attention

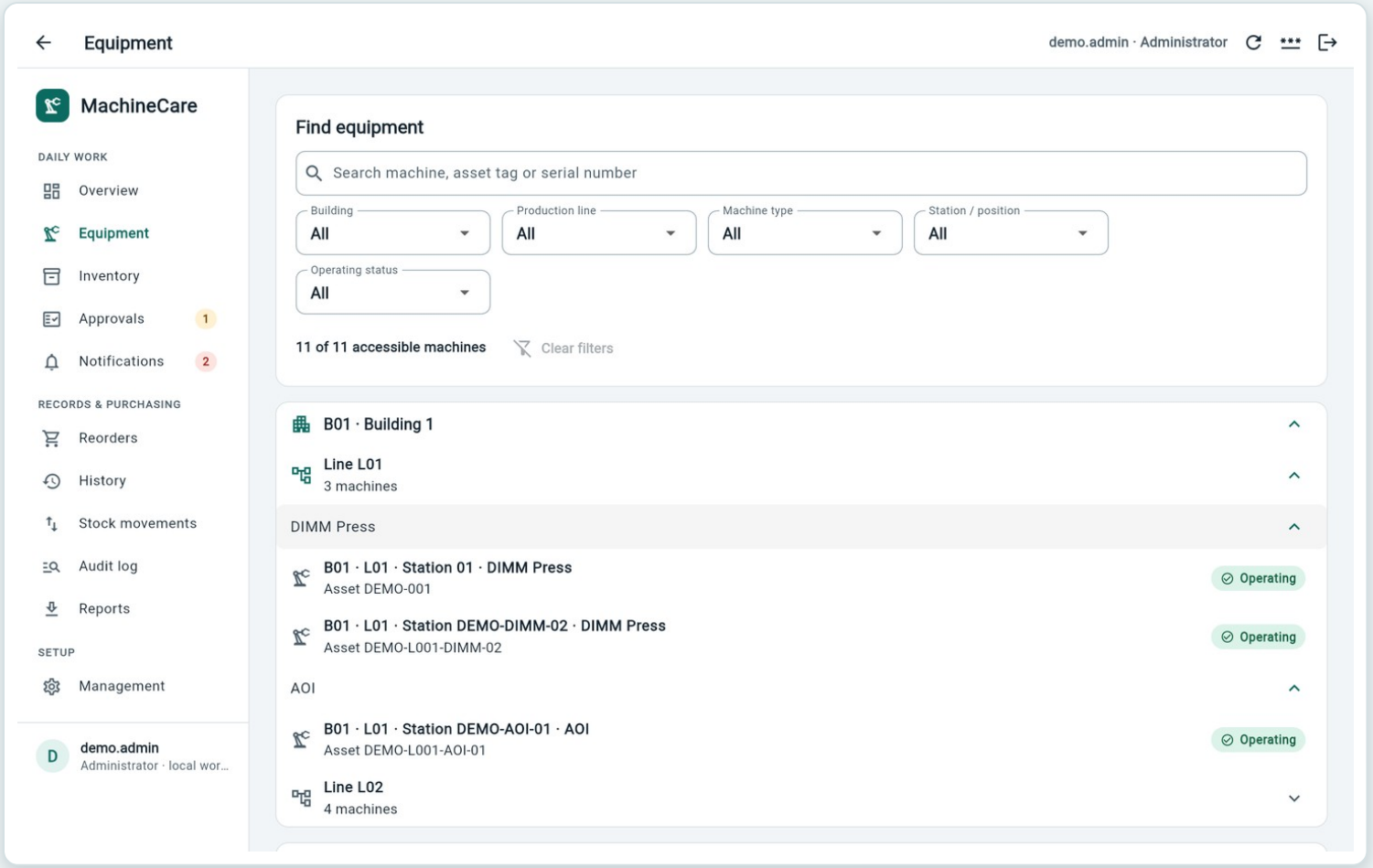
1. **Attention cards**
   Turn amber or red when action is needed; click one to jump to that page

2. **Production status**
   Operating, maintenance, stopped and retired, each with colour and icon

3. **Stock needs attention**
   Parts at or below their reorder level, with a stock bar

Demo database · illustrative values

Find the line, then the machine



Navigate the plant

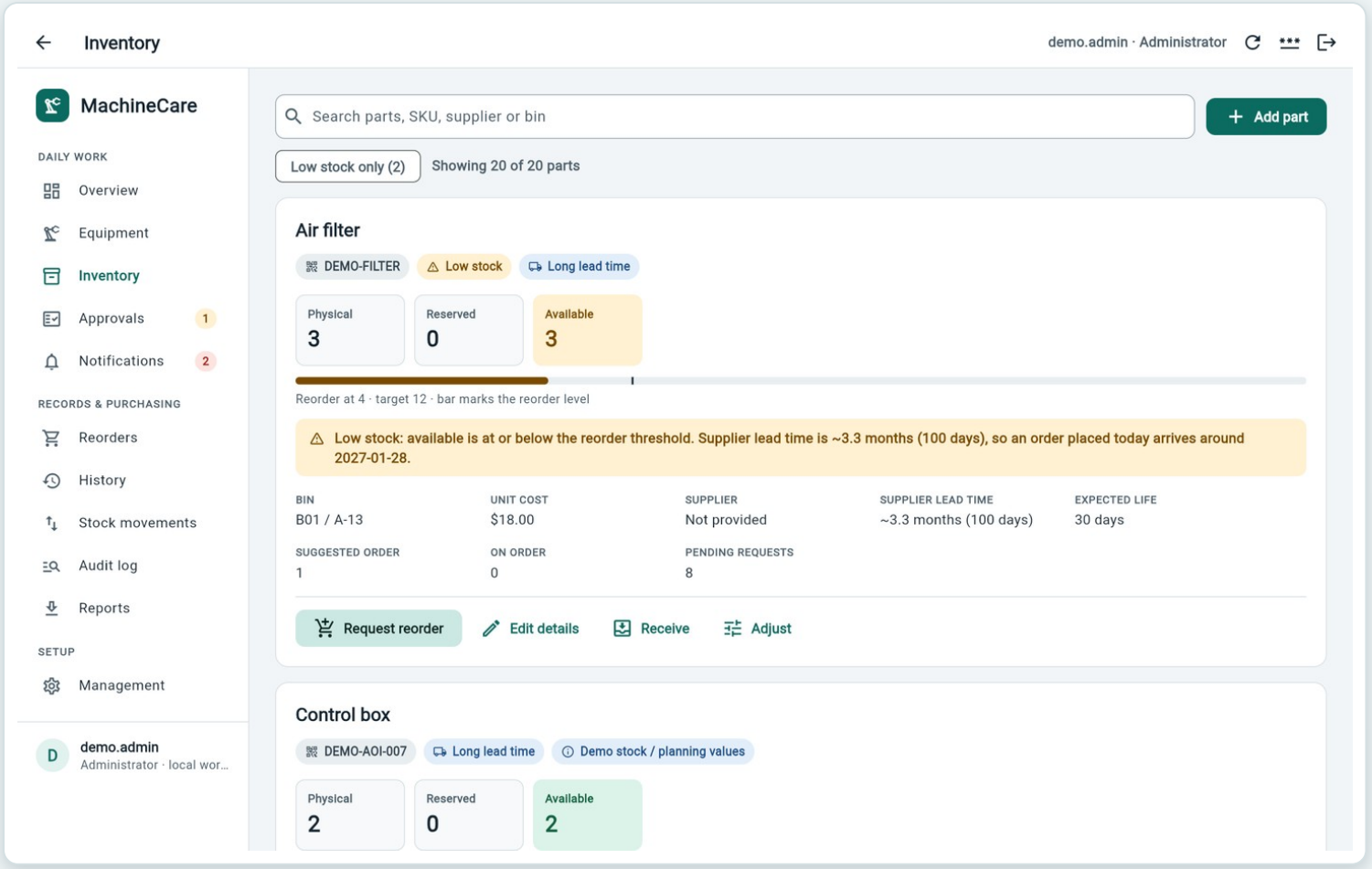
1. **Building › line › type**
   Expandable tree; each machine shows station, asset tag and status

2. **Search and filters**
   By asset tag, serial, building, line, type, station and status

3. **Scoped**
   Each user only sees the buildings and lines they are assigned

Demo database · illustrative values

Shared spare parts with honest labels



Every part shows

1. **Physical · reserved · available**
   Three clear tiles plus a bar marking the reorder level

2. **Supplier lead time**
   Shown in weeks or months; long-lead parts are flagged and arrival is estimated

3. **Real references, honest labels**
   Supplied model or serial kept; a blue tag marks demo stock and prices

Demo database · illustrative values

From request to an auditable replacement

01

Request

Technician picks machine and compatible part

02

Review

Early work needs an explanation and lead review

03

Reserve

Approval holds stock without deducting it

04

Complete

One atomic stock deduction, at completion

05

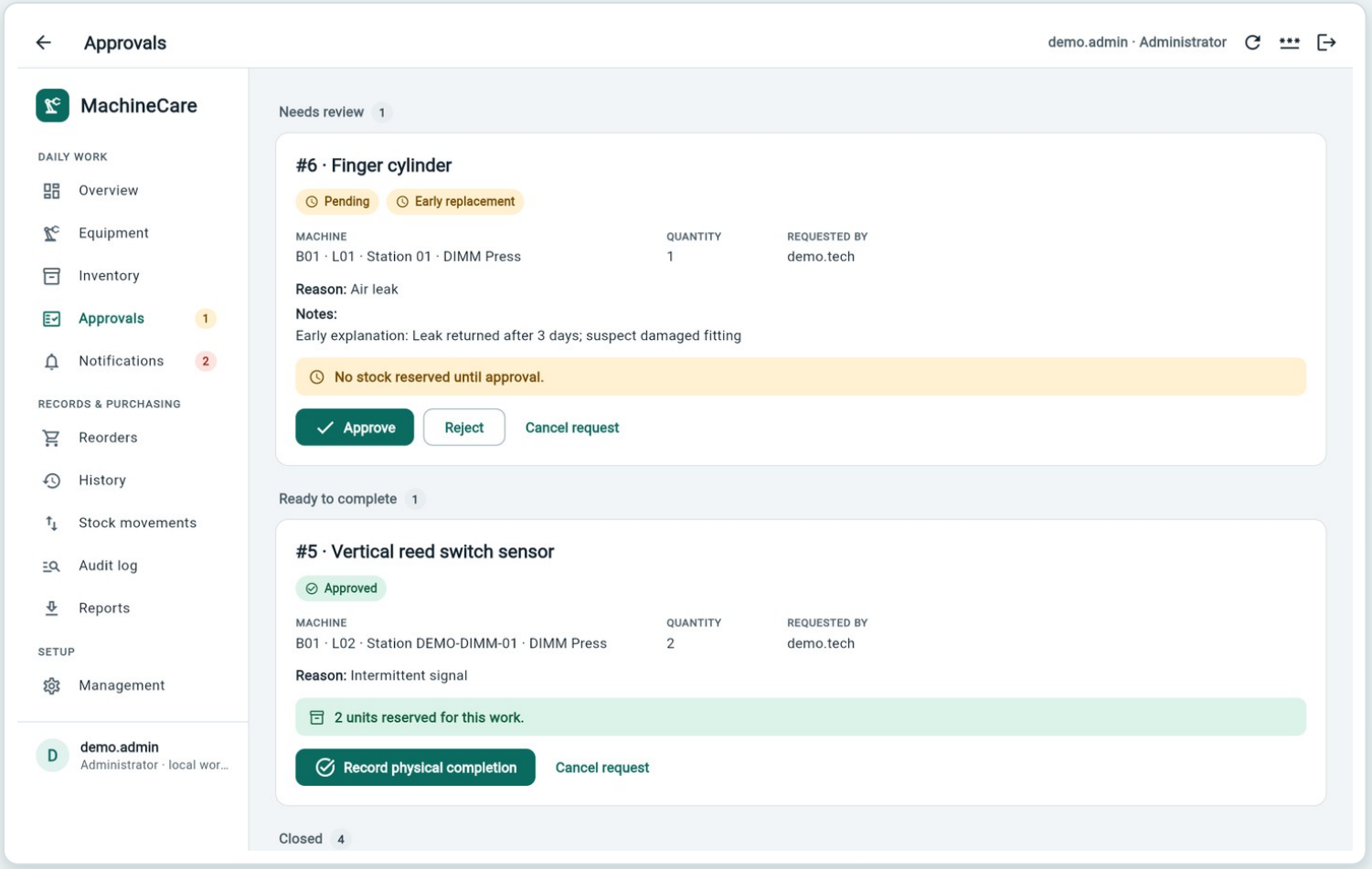
Record

History and ledger keep who, where, cost

Approval reserves stock. Only completion deducts physical inventory. Cancelling never deducts.

If someone else completes or cancels your request, you are notified. The history keeps both the original requester and the person who did the work.

Review early work before anyone records completion



Grouped by what to do next

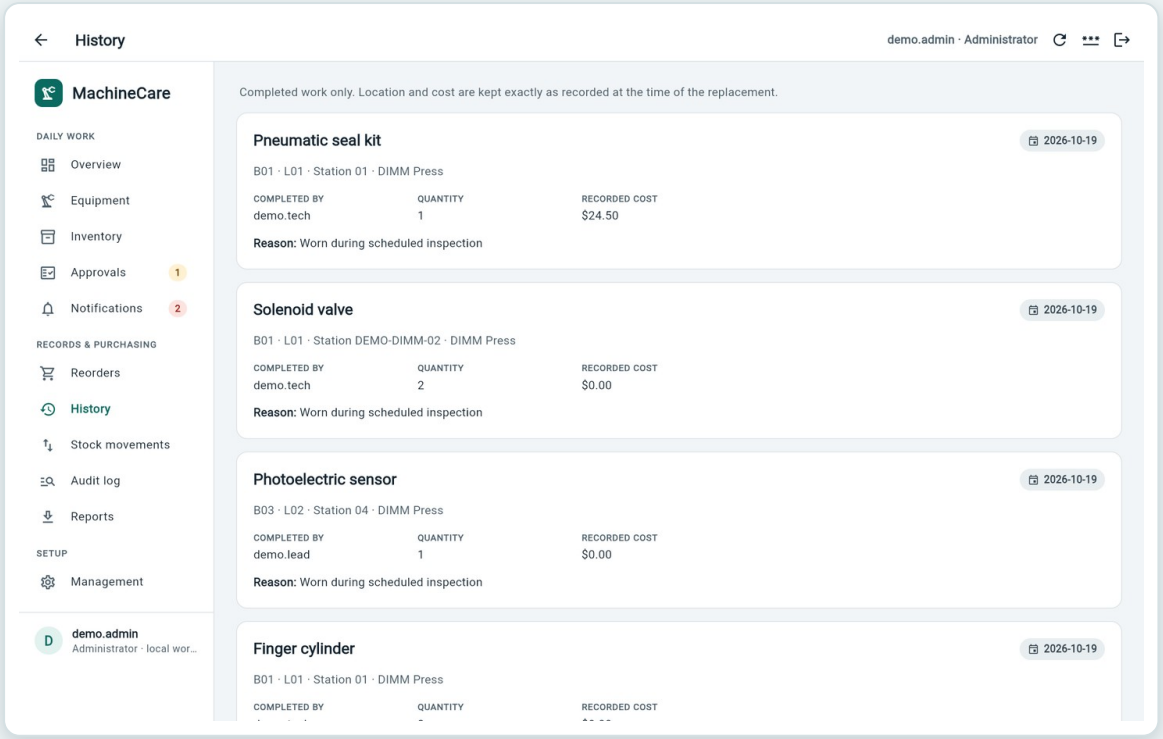
1. **#6 Pending · early**
   Finger cylinder: the technician's explanation is shown for the lead

2. **#5 Approved**
   2 reed switches reserved; ready to record physical completion

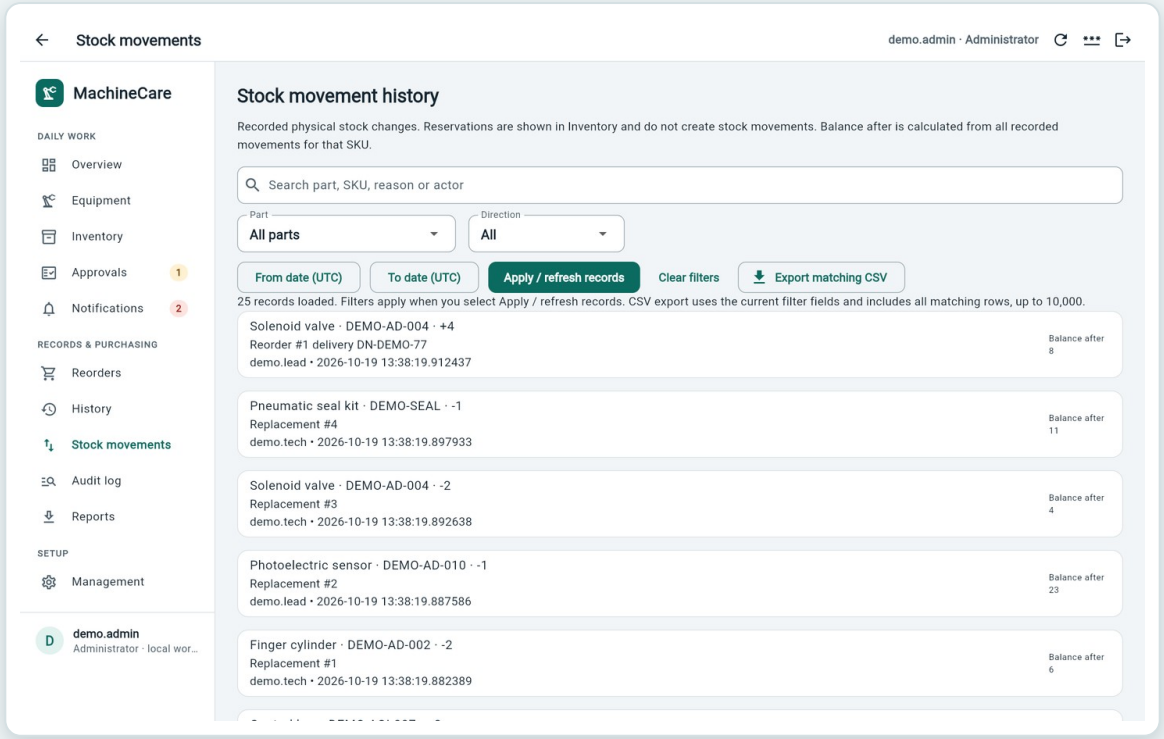
3. **#4 Completed**
   Locked; it now appears in History and Stock movements

Demo database · illustrative values

History and stock movements explain every change

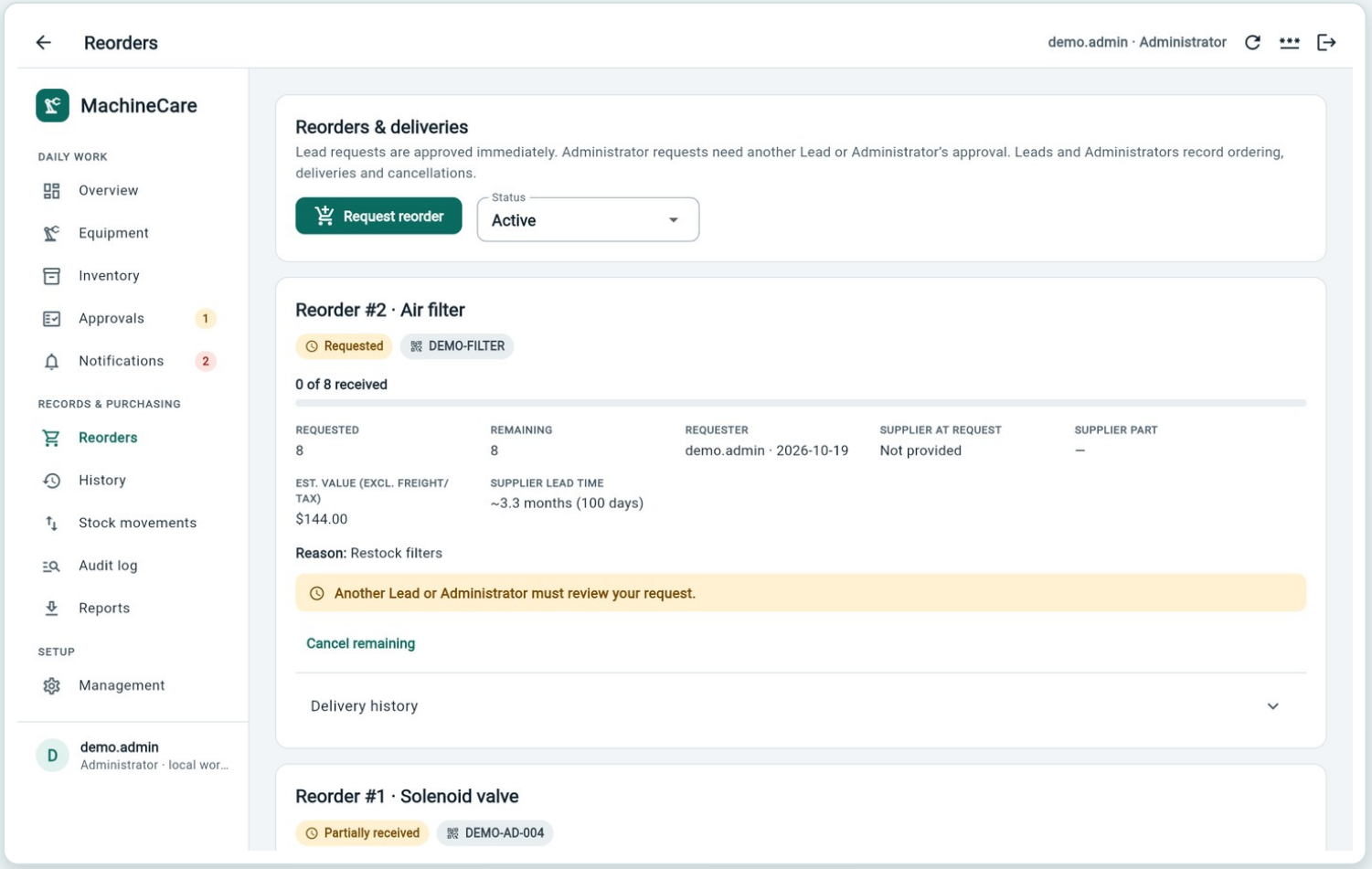


History Part, machine location, who, when, quantity and cost, kept as a snapshot even if the machine later moves or the price changes



Stock movements Every receipt, adjustment and deduction with actor, reason and running balance. Filters and CSV export

Ordering is tracked, but only receipts add stock



Request › order › receive

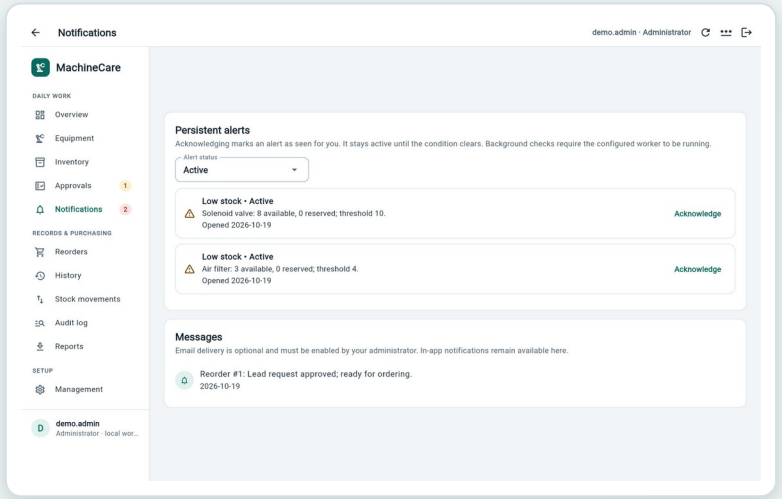
1. **Approval policy**
   Lead requests start approved; Admin requests need another manager

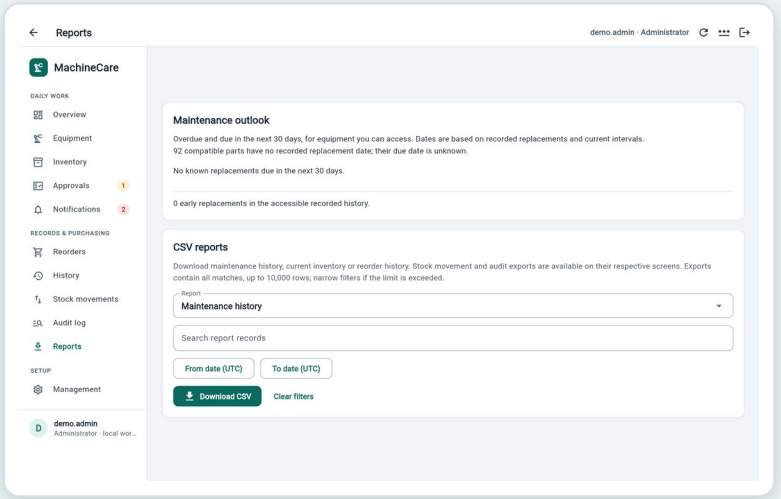
2. **Partial deliveries**
   #1 solenoid: 10 ordered, 4 received on DN-DEMO-77, 6 still open

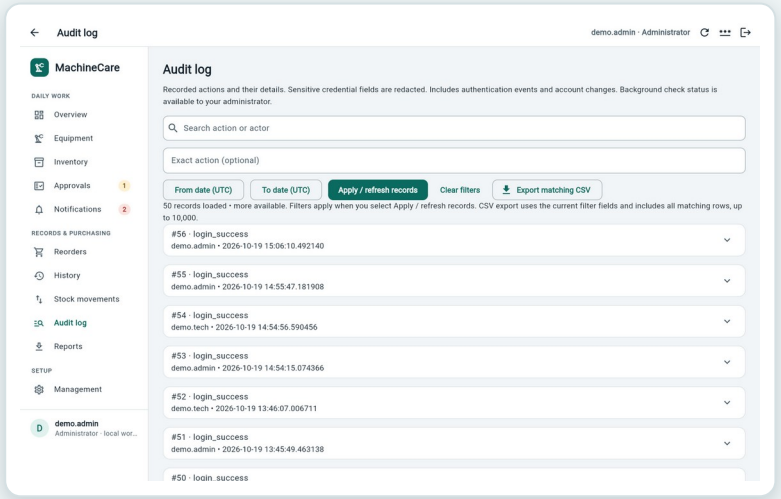
3. **Expected arrival**
   Order date + supplier lead time; late deliveries turn red. Duplicates and over-receipts are rejected

Demo database · illustrative values

Alerts, reports and a full audit trail







Persistent alerts
Low stock, shortfalls and overdue intervals stay open until the condition clears

Reports and CSV
Maintenance outlook, early-replacement counts, exports protected against spreadsheet formulas

Audit log
Logins, account changes, approvals and orders. Passwords are never recorded

BUILT IN FROM DAY ONE

Safeguards that are already working

Secure sign-in

Hashed passwords, 8-hour sessions, forced first change, lockout after repeated failures

Server-side permissions

Every action checks role and building/line scope

No double counting

Atomic transactions, duplicate-submit keys, edit-conflict detection

Safe exports

Permission checked, passwords redacted, formula injection blocked

Verified backups

Integrity-checked copies; recovery always uses a new file

Local only

Bound to this laptop (loopback); not exposed to the company network

PRODUCTION READINESS

Features are done; operations and data are not

Done	Core workflows: equipment, inventory, approvals, history, purchasing, alerts, reports, audit
Done	Role permissions, data integrity, backups and recovery tooling, automated tests
Partial	Background worker and email: built and tested, not switched on or configured
Partial	Operational data: part references real; stock, prices, bins, intervals, asset tags are demo
Missing	Production hosting: HTTPS server, company sign-in, monitoring, supported runtime
Missing	Independent security review and signed-in user acceptance for all roles
Missing	Four business rules not yet decided (multi-bin, backdating, emergency work, corrections)

WHAT'S MISSING

Release gates, owners and evidence

Gate	Still missing	Owner	Evidence of done
Operational data	Real stock counts, bins, prices, suppliers, intervals, asset tags	Maintenance lead	Signed inventory and equipment review
AOI identification	TM robot model, unknown makers, nameplate check	Maintenance lead	Verified component list
Hosting	Production web server, HTTPS, upgrade and rollback plan	IT + engineering	Reviewed staging deployment
Runtime	Python 3.10 out of support since 1 Oct 2026; Flutter SDK review	Engineering	Approved versions, tests re-run
Identity and security	SSO/MFA decision, service account, file permissions, secrets	IT / security	Independent security assessment
Backups	Scheduled, encrypted, off-host; recovery targets	IT / operations	Restore drill on a separate host
Operations	Auto-start, monitoring, logs, patching, support	IT / operations	Runbook and restart tests
User acceptance	All roles in Chrome, CSV in Excel, accessibility, load	Maintenance + eng.	Signed acceptance, no critical defects
Email (optional)	Mail server, recipients, send authorization	IT + business	Monitored test sends

DECISIONS NEEDED

Four business rules to agree before building

Multi-bin stock

If one part sits in several bins, which bin is reserved and used, and how are transfers recorded?

Backdated work

Can completion be recorded for an earlier date? How far back, in which time zone, and who approves?

Emergency work

Who can override stock or approval checks in a breakdown, and what is the deadline for review afterwards?

Corrections

How is a wrong record reversed, and what happens to stock, history and orders? Today nothing is ever deleted.

PROPOSED ROLLOUT

A staged path from demo to adoption

1 — 2 — 3 — 4

NOW

Manager review

Walkthrough on the demo copy; agree owners and success measures

NEXT

One-line pilot

Count real stock for one line; replace demo values; train users

THEN

Harden

Supported runtime, hosting design, security review, backup policy

RELEASE

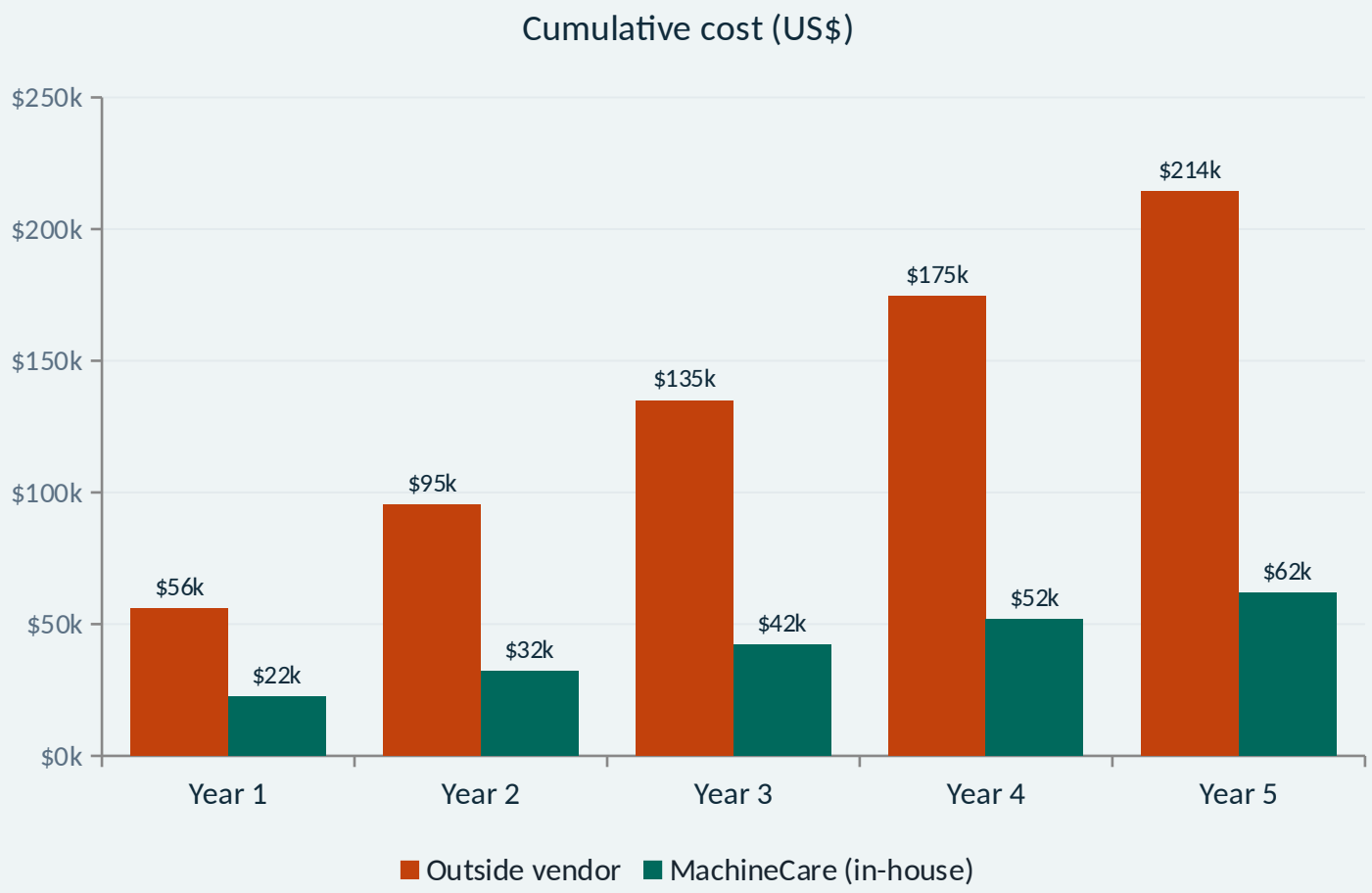
Go live

User acceptance sign-off, monitored service, support owner

Move to the next stage when the current one meets its agreed acceptance criteria. No calendar dates are promised here.

BUDGET · 5-YEAR VIEW

MachineCare costs less than a third of the outside system



Outside vendor · 5 years

$214,160

Year 1 $55,760, then $39,600 a year

MachineCare · 5 years

$61,760

Year 1 $22,400, then $9,840 a year

Saving over 5 years

$152,400

Running cost ≈ $13 vs $53 per user per month

Vendor: the quote received. Internal work on both sides at $40/h, including our in-house security team. Full model: docs/budget/MachineCare_Cost_Comparison.xlsx.

Side-by-side budget: where every dollar goes

Category	Vendor Year 1	Vendor Year 2+	Vendor 5 years	MachineCare Year 1	MachineCare Year 2+	MachineCare 5 years
Licences and fees	$47,200	$37,200	$196,000	—	—	—
Setup and deployment	—	—	—	$6,400	—	$6,400
Security (in-house team)	$800	$320	$2,080	$800	$320	$2,080
Data preparation	$3,200	—	$3,200	$3,200	—	$3,200
Training	$2,480	—	$2,480	$2,480	—	$2,480
Hosting and backups	—	—	—	$1,200	$1,200	$6,000
Running, support and administration	$2,080	$2,080	$10,400	$8,320	$8,320	$41,600
Total	**$55,760**	**$39,600**	**$214,160**	**$22,400**	**$9,840**	**$61,760**

Security is done by our own security team on both sides: 20 h initial review + 8 h a year. Data preparation and training are needed with either option, so they cancel out. The real difference is licences vs our own setup, hosting and support time.

Every individual cost line, year by year: appendix C and the Comparison sheet of the budget workbook.

What the numbers are based on

From your spreadsheet

- Vendor: $50/user/month × 62 users, $10,000 installation
- Internal labour: $40/hour
- Security review by our in-house team: 20 h ($800)
- MachineCare setup: 160 h (runtime 24, server & HTTPS 40, company sign-in 40, monitoring & backups 24, acceptance testing 32)
- Data preparation 80 h; training 1 h per person
- Hosting & backup storage $1,200/yr; support 4 h/week

Added for a fair comparison

- Our security team also reviews the vendor system: 20 h, then 8 h a year
- Yearly MachineCare security re-check: 8 h a year
- Someone still administers the vendor system: 1 h/week
- Vendor price increase: 0% a year (ask the vendor)

Without the added lines (your sheet exactly):
Vendor $201,680 vs MachineCare $60,480 over 5 years, a saving of $141,200.

Change any blue cell on the Inputs sheet; all totals recalculate. Re-export to update these slides.

THE DECISION

Approve a focused validation pilot

1

Choose one validated line

Where we count real stock and run the full workflow

2

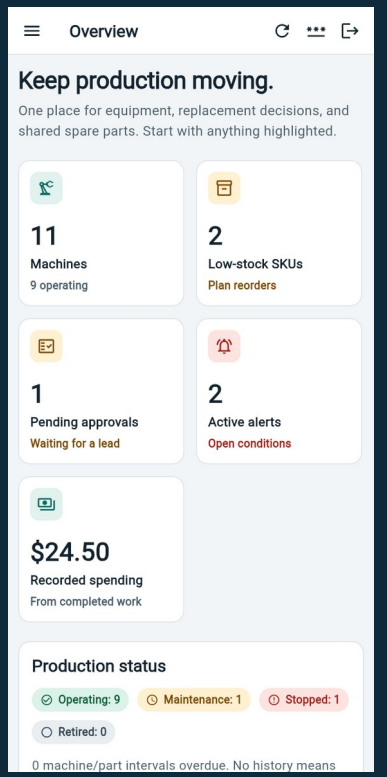
Name a data owner and an IT reviewer

Maintenance and IT/security owners for the production gates

3

Agree acceptance measures

Traceability, stock accuracy and operator usability



Also works on a phone

A working local pilot today. A controlled release after validation.

APPENDIX A

DIMM press component reference

Component	Manufacturer	Supplied identifier	Per DIMM
Load cell	Delta Transducers	Serial 2511231 (not a part number)	5
Finger cylinder	Airtac	HFD12X50	4
Vertical cylinder	Airtac	HLQ12X50S	5
Solenoid valve	Airtac	7V110-6	10
Tray cylinder	Airtac	TCL25X20S	2
Tray clamp cylinder	Unknown	HLS16X20SAS	4
Finger reed switch sensor	Airtac	DMSH A11DW	8
Vertical reed switch sensor	Airtac	D-M98W	10
Flag sensor	Panasonic	PM-T45	8
Photoelectric sensor	Panasonic	CX-441	12

APPENDIX B

AOI component reference

Component	Manufacturer	Supplied identifier	Per AOI
TM robot	TM (as supplied)	Model not supplied	1
Small light curtain	PILZ	PSEN op113H-s-30-030	4
Large light curtain	PILZ	PESN op113H-s-30-075	2
PLC PNOZ m bo.1	PILZ	Ident-772104	1
Tower light	Unknown	ASTLF-110-4-5FT-CM	2
Switching power supply	Unknown	s-120-24	1
Control box	VEVOR	Serial 240108586510146	1

APPENDIX C

Every cost line, both options

Outside vendor	Year 1	5 years
Licences	$37,200	$186,000
Installation fee	$10,000	$10,000
Setup and deployment	—	—
Security review (in-house team)	$800	$800
Yearly security re-check	—	$1,280
Physical stock count	$1,600	$1,600
Bins, prices and supplier details	$800	$800
Lead times and maintenance intervals	$800	$800
Training	$2,480	$2,480
Hosting and backups	—	—
System administration (internal)	$2,080	$10,400
Total	**$55,760**	**$214,160**

MachineCare (in-house)	Year 1	5 years
Licences	—	—
Installation fee	—	—
Runtime upgrade and re-test	$960	$960
Production server, HTTPS and deployment	$1,600	$1,600
Company sign-in (SSO)	$1,600	$1,600
Monitoring and scheduled backups	$960	$960
User acceptance testing	$1,280	$1,280
Security review (in-house team)	$800	$800
Yearly security re-check	—	$1,280
Physical stock count	$1,600	$1,600
Bins, prices and supplier details	$800	$800
Lead times and maintenance intervals	$800	$800
Training	$2,480	$2,480
Server / VM hosting	$900	$4,500
Off-host backup storage	$300	$1,500
Patching and monitoring	$3,120	$15,600
User help and fixes	$3,120	$15,600
Small changes and improvements	$2,080	$10,400
Total	**$22,400**	**$61,760**